

S26 – Data Pipeline 1 – Project: XML-Based Museum and Cultural Heritage Management Platform

This project focuses on designing and implementing an XML-based database platform for a museum and cultural heritage institution. The system will integrate and manage diverse entities—such as artifacts, collections, exhibitions, artists, historical periods, cultural sites, curators, loans, restoration projects, and visitors—to ensure efficient preservation, organization, and access to cultural heritage information. The project will cover:

- Data modelling and management: Structuring and maintaining information about entities and their relationships, including artifact descriptions, collection inventories, exhibition schedules, artist biographies, provenance records, restoration histories, loan agreements, and cultural heritage classifications.
- User-oriented functionalities: Exploiting the database to support various user needs, such as searching for artifacts or exhibitions, browsing collections by historical period or cultural origin, tracking loans and restoration activities, managing exhibition schedules, and generating reports on collections, visitor attendance, and preservation activities.

The platform will showcase the effective exploitation of XML technologies for real-world museum and cultural heritage collection management.

- Write an **XML Schema** that represents your modeling
- Write an **XML database** that is valid against this model, containing a representative extract of the database.
- Imagine **6 different scenarios** requiring the visualization of part of this data and/or new data computed from it and write **6 XSL stylesheets** to implement them. (visualisation meaning outputting HTML data)
- Imagine **2 additional scenarios** requiring the exploitation of part of the **XML database** in another XML format and write an XSL stylesheet for this.
- Imagine **1 additional scenarios** requiring the exploitation of part of the XML database in **JSON format** and write an **XSL stylesheet** for this. Write a **JSON schema** for the output format.
- Imagine **1 additional scenarios** requiring the exploitation of part of the XML database in **YAML format** and write an XSL stylesheet for this.
- Write a Python or Java program to **load an XML file**, **parse it**, **validate** it against the XML schema and **apply an XSL stylesheet** (using XML APIs).
- Optionally, provide a second implementation of one of your scenarios in Python or Java (using XML APIs, e.g. DOM).
- All the **xml/xsd/xsl/json/yaml, java and python codes** should be **commented**.

Report:

Write a report (pdf format) containing the following: indicate **how you distributed the work load in the group** (overall **percentage** of contribution to the project for each student and **precise list of tasks** to which each member contributed

Indicate your **working environment**, and the tools or online services that you used.

Describe the **principles of your models and explain and discuss (advantages/disadvantages)** your modeling choices. Choose one modeling problem that you addressed and explain the solution that you chose. Nb: I do **NOT expect here a general discussion on the XML language**.

Describe the scenarios for which you implement a solution. Choose **one scenario** requiring quite a **complex** solution and explain it.

Additionally, in case you make use of chatGPT, copilot or any other AI assistant at some point in the project, write an additional dedicated section where you indicate the precise prompts that you used, what you got, what you corrected, etc.

The report should not exceed **6 pages** (optional section on AI assistance excluded).

You will be evaluated according to the quality and richness of your modeling (modular schema, fine grained type definitions and constraints), the quality and richness of the **addressed scenarios/use cases/functionalities, the quality of your stylesheets (prefer a recursive XSL programming style that extensively uses XPath)**, the quality of your Java and Python codes, and the quality of your report showing your ability to take a step back.

Submission:

Upload a zip file containing:

- your **report** (in pdf format) including list of tasks each member completed
- your **source XML file**
- your **XML Schema** file
- your **XSLT files** containing in comments their description in natural language
- the **outputs of your XSLT** transformations
- the **JSON Schema** file for the **JSON output** format
- your **Python and/or Java source files**